

Soothing Systems: A Meditative Somaesthetic Experience

Samuel Shields

Computational Media, University of California, Santa Cruz
samshiel@ucsc.edu

Edward F. Melcer

Computational Media, University of California, Santa Cruz
emelcer@ucsc.edu

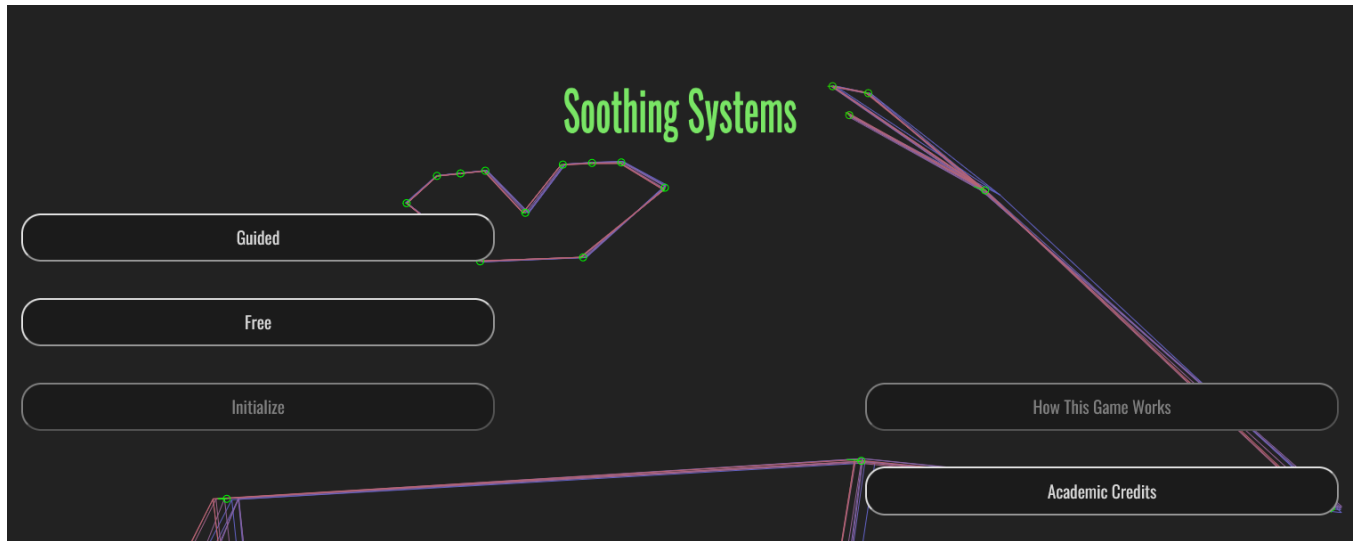


Figure 1: The title screen for Soothing Systems. Body tracking of the user is performed by a consumer-grade webcam and displayed on screen as a simplified skeletal outline.

ABSTRACT

We present Soothing Systems, a web-accessible video game that utilizes body tracking through a webcam to guide users through yoga-inspired poses for a meditative somaesthetic experience). The overall experience consists of enacting different meditative poses by presenting the user targets that can be hit with specific body parts represented by their avatar. Visual and audio guides are provided for breathwork practices, and an on-screen narrator prompts users to reflect as they explore the game world presented on screen. Through this open-ended exploration of movement on the screen, users leave traces of movement on a canvas, leaving an abstract, colorful artwork representation, which can be printed out and taken away with the user as a souvenir. Through embodied explorations of a 2D game space, we aim to provide a space for meditation and relaxation that contributes to an ongoing dialogue of explorative somaesthetic experiences in the HCI community.

CCS CONCEPTS

• **Human-Centered Computing**; • **Human Computer Interaction (HCI)**; • **Interaction Paradigms**; • **Web-Based Interaction**;

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3648651>

KEYWORDS

Somaesthetic Design, Web Application, Augmented Reality, Embodiment

ACM Reference Format:

Samuel Shields and Edward F. Melcer. 2024. Soothing Systems: A Meditative Somaesthetic Experience. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3613905.3648651>

1 INTRODUCTION

Embodiment and soma-based design has increasingly become core to HCI scholarship as more and more applications, games, and digital interfaces are designed around a person's mind being deeply embedded in a body instead of mind and body existing as separate entities [4]. This style of design often leans on using unique interfaces that emphasize mimetic controls that provide more direct and representative modes of play - for example, swinging a controller shaped like a wand to cast a fishing reel in a video game. It is observed that this avenue of control, which uses imitation of a task instead of simulation, can produce unique outcomes in users [2]. For instance, a strong view of embodiment in educational settings asserts that most, if not all learning is ultimately embodied in some dimension [1].

Soma-driven design is a relatively new framework to understand how to create digital experiences that extend past a 2D screen, holding digital interfaces as extensions of user's real bodies [7]. It

emphasizes first person perspectives as a foundation for design, as it requires the designer to identify phenomena from an experiential perspective across multiple interaction domains rather than starting in a well-defined purely digital space [7]. This style of design also emphasizes movement and engagement across different modalities of sense information - it pays careful consideration to what sensations a user will encounter as part of their digital experience [15].

In game design, it is observed that video games produce novel forms of embodiment that are unique to the interface and simulation being presented to a player [4]. These forms of embodiment have the potential to produce strong learning effects in users [4]. Embodiment is also increasingly seen as a therapeutic avenue, providing novel ways to interact not only with physical disabilities but also with mental disorders as well [13]. With these concepts in mind, we introduce Soothing.Systems, a web-application video game that aims to use body-tracking libraries and a consumer-grade webcam to create a meditative environment for end-users on commonly available hardware. Soothing.Systems provides a free-form interface for users to attempt yoga-inspired poses while receiving feedback based on a variety of design strategies derived from somaesthetics. By presenting Soothing.Systems as a tool that can be run as a simple web application without additional hardware beyond a web camera, we hope to provide a meditative environment at low or no cost and without complicated hardware setup (such as virtual reality headsets or additional connected devices). The core concept of the video game presents an avatar representation of the player in a black background, with a narrator and targets presented to enable the player to playfully generate new procedurally generated patterns behind the avatar. The player is given an option to screenshot and print out any produced designs, giving a “souvenir” for the player to remember their experience.

2 SYSTEM DESIGN AND FEATURES

Soothing.Systems main goal is to have players explore yoga-inspired poses in a meditative fashion by providing an interface that employs various forms of visual and audio feedback. It consists of a guided play mode where a player’s avatar is represented on screen based on information provided by their computer’s webcam. Players are then invited by a narrator to try interacting with targets, which roughly correspond to a yoga-inspired pose. For example, the first pose presented is a simple “sun breath”, where users bring their outstretched arms over their head and back down again. As the player interacts with the system, different visual elements of the avatar are saved to a background canvas, creating a visual history of the player’s movements on screen. A configuration menu is available for players to change what effects are rendered on their avatar - displaying a parameterized list of game features that can be turned on or off to customize what visuals the player will see (e.g., different colors for painting, breathing trackers, and so on).

2.1 Body Position Tracking and Targets

Soothing.Systems was built with consumer-grade computer hardware in mind - interacting with the system should only require a computer with a webcam of sufficient resolution. This goal was achieved by leveraging several open source libraries - P5.js is used

for player rendering and gameplay elements, Tone.js is used for sound generation, and MediaPipe was used for body and pose tracking. MediaPipe in particular was selected for pose tracking over other libraries as it has been found to be the most accurate for estimating Yoga poses a user presents [9]. The combination of these technologies results in an application that can run without a backend and inside of a web browser. MediaPipe provides pose information to P5.js, which is then encoded to represent specific body parts that can interact with the game world. This allows us to present “targets” on screen for players to “touch” by moving their body to reposition their avatar on screen. Figure 2 shows what a target interaction looks like. Targets have feedback in the form of particle generation effects tied to it, erupting in a confetti of colorful circles while also playing a pleasing set of tones. This effect is inspired by work in “GameFeel”, which aims to enhance clarity of feedback and bring unique physicality to the game world [10].

2.2 Narrator

During gameplay, an onscreen prompt appears that instructs the player to reflect on their experience and encourages them to explore different mechanics in the game. For example, the narrator might invite the player to synchronize their breath to an in-game prompt, or to try and interact with targets using a specific body part. This mechanic helps provide more in-depth instruction on how to use the game, and has a “voice” in the form of a dynamically changing synthesized tone that activates when displaying new text to the user. We portray these prompts as coming from a Narrator, providing a guiding companion to the end-user as they explore their poses [14]. Figure 3 shows how the narrator appears on screen visually and provides an example instruction that it provides.

2.3 Breathing Guides

Utilizing the soma-based framework, a key element of meditative experiences often revolves around self-regulation of the breath [11, 16]. As such, Soothing.Systems provides breathing guides in its interface for users, which give a static rhythm for users to try and align their breath to. We provide two mechanics for this: First, a visual effect which outlines the users’ body onscreen, slowly shifting from white to black in a recurring cadence. Second, a sweeping synthesizer tone that oscillates back and forward at a similar cadence. These provide guidelines for the user to synchronize their breath during their experience with the game. The in-game narrator provides instructions on how to use these different mechanics during gameplay. Figure 4 illustrates how the silhouette visually changes to represent in and out breaths.

2.4 Soundscapes

The library Tone.JS is used to generate all sounds in the game based on simulated analog synthesizers. This is done to create a coherent soundscape, and provide cues for critical in game actions mentioned in previous subsections. I.e., the soundscape provides target feedback, narrator, and breath sound effects, all of which help the player situate themselves in the gameworld. For example, when a user successfully touches a target, a “twinkling” sound composed of major scale tones is generated to indicate a successful hit. The narrator,



Figure 2: A screenshot of a hand interacting with a target, causing colorful particles to fill the screen.

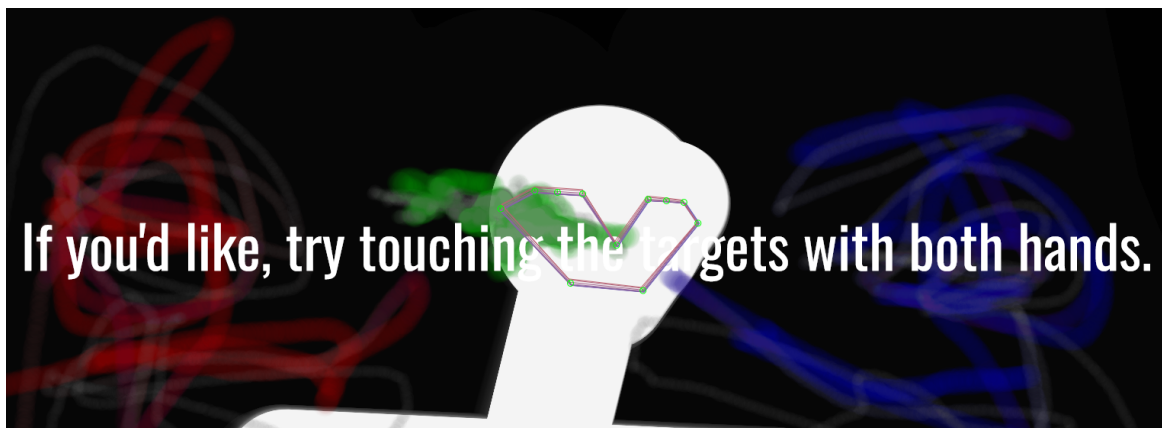


Figure 3: The narrator instructing the user to interact with the targets on screen.

inspired by the audio of character dialogue from Nintendo's *Animal Crossing* (Animalese), speaks in an incomprehensible tone that nonetheless sounds like a narrator is presenting verbal guidance and reassurance to the user. The breath tracker provides a sweeping motion, giving a rhythmic guidance that a user can use to track their own breath. Outside of these sounds, an ongoing white noise

generator is used to help focus the end user on the game experience, as this style of noise has been found to augment cognitive and perceptual tasks in digital environments [6]. This soundscape is constantly yet subtly changing, providing insulation from outside sounds while also promoting focus on in-game actions.

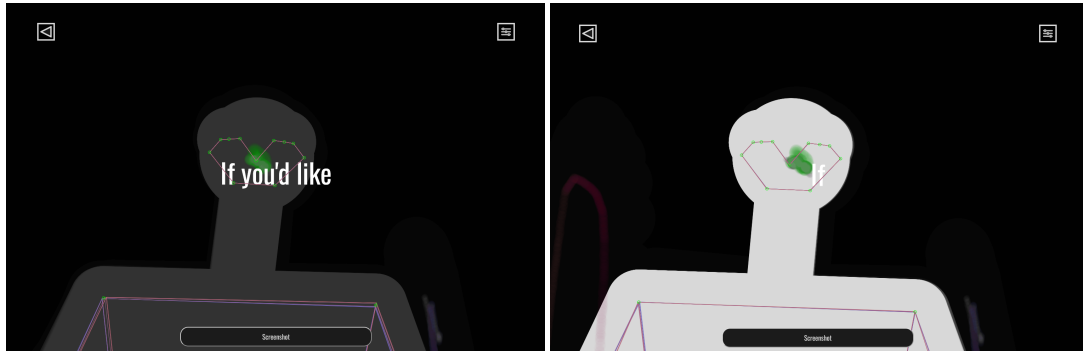


Figure 4: A demonstration of a visual breathing guide. A silhouette changes from dark to light in a rhythmic fashion, providing a visual guide for the user to sync their breath to.

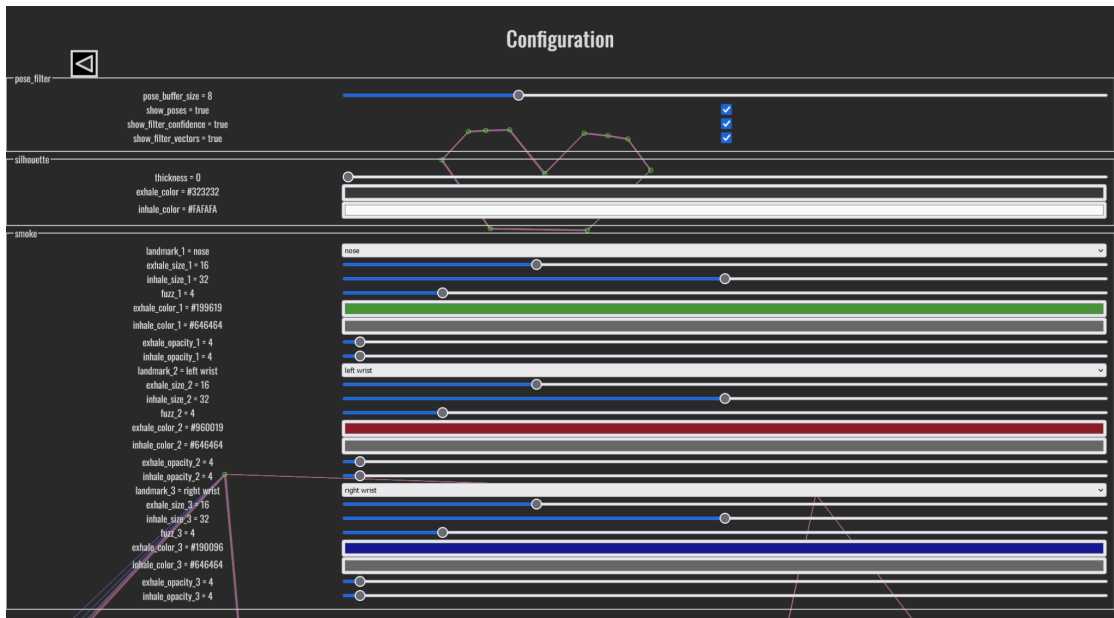


Figure 5: The configuration panel for Soothing Systems. Colors, drawing sizes, and more can be changed to provide a unique experience for each user.

2.5 Procedural Souvenir Generation

As the player moves throughout the game, different visual effects are drawn to a non-refreshing canvas, leaving a trail of in-game actions reflected back to the player. For example, player limbs might leave colored smoke trails behind as they move around, leaving an artistic trace of movement throughout the game. These accumulate throughout the play session, letting the player to either focus on achieving in-game objectives (e.g., interacting with targets) or instead choose to playfully “draw” new abstract artworks with their body positioning over time. This effectively creates a digital souvenir, inspired by “data souvenirs” used to represent patron interactions at a museum [12], and provides a touchstone for users to return to after they leave the installation space. Users can be given the option to print out their souvenir or take a picture of it to take

with them afterwards. Souvenirs are aimed to be unique user to user, and as such leverage a configuration menu to provide unique visual effects to each user. This configuration menu can be seen in Figure 5. These tangible forms of souvenirs have been shown to be effective motivators in educational and health domains [5, 8], and Soothing.Systems similarly leverages this technique for a unique motivational reward.

3 INSTALLATION SETUP AND USAGE

Setup for Soothing Systems consists of a display connected to a laptop running the application through a web camera. Every new user will start with a new session, refreshing the web page. The facilitator will grant necessary permissions to turn on the computer’s camera, and basic instruction will be given to the user to position themselves approximately 6 feet away from the camera to have

their entire body represented within the game space. The display will be facing the user, and a pair of headphones will be provided to the user for audio. Users may take screenshots at any point during the presentation, which can then be printed using a color printer connected to the computer. 1-3 poses will be presented to the user, though the user is free to explore creating art or interacting with the application in whichever way they see fit.

4 CONCLUSION

Soothing Systems is a web application meant to explore connection between video gameplay and embodiment while offering a soothing relieve. By utilizing multiple senses and leveraging the body as a controller, we aim for this somaesthetic experience to provide a sense of calm and a space to explore breath and movement on the conference floor. We aim to make the experience memorable by offering a souvenir of usage during or after the installation, and leverage a configuration menu to make the experience unique to each user.

REFERENCES

- [1] Abrahamson, D., & Lindgren, R. (2014). Embodiment and Embodied Design. <https://doi.org/10.1017/cbo9781139519526.022>
- [2] de Castell, S., Jenson, J., & Thumlert, K. (2014). From Simulation to Imitation: Controllers, Corporeality, and Mimetic Play. *Simulation & Gaming*, 45(3), 332–355. <https://doi.org/10.1177/1046878114542316>
- [3] Farr, W., Price, S., & Jewitt, C. (2012). An introduction to embodiment and digital technology research: Interdisciplinary themes and perspectives [Working Paper]. NCRM. <https://eprints.ncrm.ac.uk/id/eprint/2257/>
- [4] Farrow, R., & Iacovides, I. (2014). Gaming and the limits of digital embodiment. *Philosophy & Technology*, 27(2), 221–233. <https://doi.org/10.1007/s13347-013-0111-1>
- [5] Grasse, K. M., & Melcer, E. F. (2020, November). Generation: A novel fabrication game for simulating evolution and natural selection. In *Extended Abstracts of the 2020 Annual Symposium on Computer-Human Interaction in Play* (pp. 38–42).
- [6] Herweg, N. A., & Bunzeck, N. (2015). Differential effects of white noise in cognitive and perceptual tasks. *Frontiers in Psychology*, 6. <https://www.frontiersin.org/articles/10.3389/fpsyg.2015.01639>
- [7] Höök, K., Caramiaux, B., Erkut, C., Forlizzi, J., Hajinejad, N., Haller, M., ... & Tobiasson, H. (2018, February). Embracing first-person perspectives in soma-based design. In *Informatics* (Vol. 5, No. 1, p. 8). MDPI.
- [8] Khot, R. A., Aggarwal, D., Pennings, R., Hjorth, L., & Mueller, F. F. (2017, May). Edipulse: investigating a playful approach to self-monitoring through 3D printed chocolate treats. In *Proceedings of the 2017 CHI conference on human factors in computing systems* (pp. 6593–6607).
- [9] Kishore DM, Bindu S, Manjunath NK. Estimation of Yoga Postures Using Machine Learning Techniques. *Int J Yoga*. 2022 May-Aug;15(2):137-143. doi: 10.4103/ijoy.ijoy_97_22. Epub 2022 Sep 5. PMID: 36329766; PMCID: PMC9623892.
- [10] M. Pichlmair and M. Johansen, "Designing Game Feel: A Survey," in *IEEE Transactions on Games*, vol. 14, no. 2, pp. 138–152, June 2022, doi: 10.1109/TG.2021.3072241.
- [11] Morris, C., Liu, P., Riecke, B. E., & Maes, P. (2023). InExChange: Fostering Genuine Social Connection through Embodied Breath Sharing in Mixed Reality. *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*, 1–5. <https://doi.org/10.1145/3544549.3583917>
- [12] Petrelli, D., Marshall, M. T., O'Brien, S., McEntaggart, P., & Gwilt, I. (2017). Tangible data souvenirs as a bridge between a physical museum visit and online digital experience. *Personal and Ubiquitous Computing*, 21(2), 281–295. <https://doi.org/10.1007/s00779-016-0993-x>
- [13] Rohricht, F., Gallagher, S., Geuter, U., & Hutto, D. D. (2014). Embodied cognition and body psychotherapy: The construction of new therapeutic environments. *Sensoria: A Journal of Mind, Brain and Culture*, 10(1), 11. <https://doi.org/10.7790/sa.v10i1.389>
- [14] Trocchianesi, R., Pinardi, D., & De Marco, A. (2011). Design and Narration: Languages, structures and narrative models as matter for design. *Strategic Design Research Journal*, 4(2), 59.
- [15] Tsaknaki, V., Balaam, M., Ståhl, A., Sanches, P., Windlin, C., Karpashevich, P., & Höök, K. (2019). Teaching Soma Design. *Proceedings of the 2019 on Designing Interactive Systems Conference*, 1237–1249. <https://doi.org/10.1145/3322276.3322327>
- [16] Zhu B, Hedman A, Feng S, Li H, Osika W, Designing, Prototyping and Evaluating Digital Mindfulness Applications: A Case Study of Mindful Breathing for Stress Reduction, *J Med Internet, Res* 2017;19(6):e197, URL: <https://www.jmir.org/2017/6/e197>, DOI: 10.2196/jmir.6955